

Random Signal Theory–EEE 554

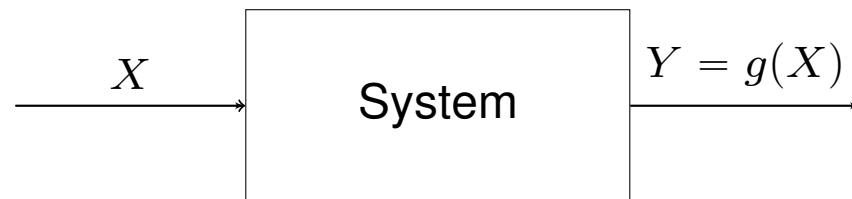
Lecture Notes 2

Transformations of a Random Variable

[Reading: Kay, Sections 5.7, 10.7]

Transformation of one RV

Obtain RV Y by transforming RV X using $Y = g(X)$



Such transformations common in systems processing: output is transformed input

Example:

A signal $s[n]$ is transmitted over a noisy channel; signal received is $y[n] = s[n] + x[n]$

At time $n = n_1$, $Y = y[n_1]$ is a RV obtained by transforming the Gaussian RV $X = x[n_1]$

Transformation function is $Y = g(X)$, since $y[n_1] = g(x[n_1]) = s[n_1] + x[n_1]$

Different cases of RV Y , depending on the type of function $g(X)$

Case 1. $g(x)$ is a one-to-one function (monotonically increases or decreases)

$\Rightarrow g(x)$ is differentiable and invertible, i.e., $x = g^{-1}(y)$ is the only solution to $y = g(x)$

$\Rightarrow$ sample space of Y : $\{y \in S_Y : y = g(x), x \in S_X\}$

CDF:
$$F_Y(y) = F_X(g^{-1}(y))$$

and PDF (or PMF for discrete):
$$f_Y(y) = \frac{1}{|g'(g^{-1}(y))|} f_X(g^{-1}(y))$$

where $g'(x) = \frac{d}{dx}g(x)$ and

$$\frac{d}{dy}g^{-1}(y) = \frac{1}{g'(g^{-1}(y))} \quad \text{[see Kay, Equation (10.30), page 313]}$$

Examples of Case 1: **[Kay, Section 5.7, 10.7]**

► Linear transformation, X DRV and $Y = aX + b$ **[Kay, Example 5.4]**

► Linear transformation, X CRV and $Y = aX + b$ **[Kay, Example 10.5, $X \sim \mathcal{N}(0, 1)$]**

$$\text{Given } F_X(x), F_Y(y) = F_X\left(\frac{y-b}{a}\right) \text{ and } f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

► Exponential transformation, $Y = \exp(X)$ **[Kay, Example 10.6, $X \sim \mathcal{N}(0, 1)$]**

Example of Case 1: linear transformation

Let $X \sim \mathcal{U}(0, 2)$ and $Y = g(X) = 3X + 1$. Find the CDF $F_Y(y)$ of Y

$$F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = \Pr(X \in R_y) = F_X\left(\frac{y-1}{3}\right)$$

R_y is the region in S_x obtained as $R_y = \{x \in S_X : g(x) = (3x + 1) \leq y, y \in S_Y\}$

Since sample space of X is $S_x \in \{0, 2\}$, then $S_Y \in \{y_{\min}, y_{\max}\}$

where, using monotonic transformation $y = g(x) = 3x + 1$, $y_{\min} = 1$ and $y_{\max} = 7$

Since $X \sim \mathcal{U}(0, 2)$, its CDF is $F_X(x) = \begin{cases} 0, & x < 0 \\ 0.5x, & 0 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$

Y is also uniform, $Y \sim \mathcal{U}(1, 7)$, since its CDF is

$$F_Y(y) = F_X\left(\frac{y-1}{3}\right) = \begin{cases} 0, & \frac{y-1}{3} < 0 \\ 0.5 \frac{y-1}{3}, & 0 \leq \frac{y-1}{3} < 2 \\ 1, & \frac{y-1}{3} \geq 2 \end{cases} = \begin{cases} 0, & y < 1 \\ \frac{y-1}{6}, & 1 \leq y < 7 \\ 1, & y \geq 7 \end{cases}$$

The PDF of Y , $f_Y(y)$, can be obtain by taking the derivative of $F_Y(y)$

General result of linear transformation of uniform RV

if $X \sim \mathcal{U}(x_1, x_2)$ and $Y = aX + b$, $a > 0$, then $Y \sim \mathcal{U}(ax_1 + b, ax_2 + b)$

Case 2. $g(x)$ is a **many-to-one function**

$\Rightarrow g(x)$ has multiple roots, more than one value of x provides the same value of y

if $Y = g(X)$ and $x_i = g^{-1}(y)$, $i = 1, 2, \dots, N$ (N roots), then PDF of Y is

$$f_Y(y) = \sum_{i=1}^N \frac{1}{|g'(x_i)|} f_X(x_i)$$

Examples of Case 2

- ▶ $Y = X^2$, obtain PDF directly **[Kay, Examples 5.5, 5.6, 10.7]**
- ▶ $Y = X^2$, obtain CDF, take derivative of CDF to obtain PDF **[Kay, Example 10.8]**

Example:

Find PDF of $Y = g(X) = |X|$, given CDF $F_X(x)$ of X and sample space $S_X \in \mathbb{R}$

Since $y = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0 \end{cases}$, then sample space of Y is $S_Y \in \mathbb{R}^+$

Also, $y = |x|$ has two roots; for $x_0 > 0$, both x_0 and $-x_0$ yield the same value of y

Find the CDF of Y ,

$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) = \Pr(g(X) \leq y) = \Pr(|X| \leq y) \\ &= \Pr(-y \leq X \leq y) = F_X(y) - F_X(-y), \quad y \geq 0 \end{aligned}$$

To find the PDF, take the derivative of the CDF

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} (F_X(y) - F_X(-y)) \\ &= f_X(y) - (-)f_X(-y) = f_X(y) + f_X(-y), \quad y \geq 0 \end{aligned}$$

For $X \sim \mathcal{U}(-10, 10)$, then $S_X \in \{-10, 10\}$, since $Y = |X|$, then $S_Y \in \{0, 10\}$

Using PDF of X , **[Kay, Table 11.1]**

$$f_X(y) = \frac{1}{20}, -10 < y < 10$$

$$f_X(-y) = \frac{1}{20}, -10 < -y < 10 \text{ or } -10 < y < 10$$

$$\begin{aligned} f_Y(y) &= f_X(y) + f_X(-y), y \geq 0 \\ &= \frac{1}{20} + \frac{1}{20} = 0.1, 0 < y < 10 \end{aligned}$$

So Y is a uniform RV $\Rightarrow Y \sim \mathcal{U}(0, 10)$

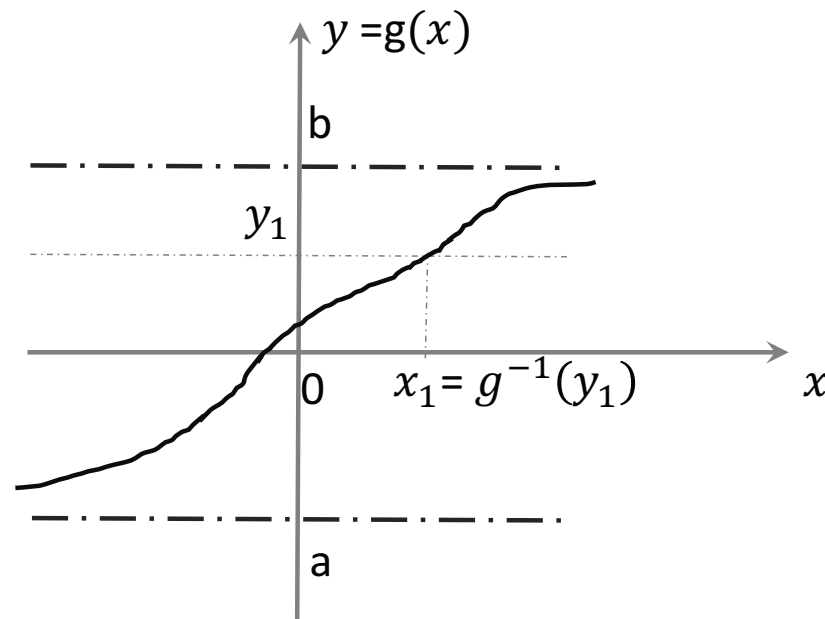
General case of $Y = g(X)$ (Case 1 or Case 2)

find PDF as the derivative of the CDF [Kay, Example 10.8]

Given $F_X(x)$ and $S_X \in \mathbb{R}$, and assuming that $S_Y \in (a, b)$, for $a, b \in \mathbb{R}$

then the CDF of Y is $F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = \Pr(X \in R_y)$

where R_y contains all points in $x \in S_x$ such that $g(x) \leq y$ or $x \leq g^{-1}(y)$, for $g(x)$ invertible



All possible such regions must be taken into consideration.

For example, from figure for $y = g(x)$, then R_y consists of three separate regions:

Region 1: $y \geq b$, Region 2: $y < a$, Region 3: $a \leq y < b$.

For Region 1: $y \geq b$,

$F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = 1$ since all values in S_X are below $y = b$

For Region 2: $y < a$,

$F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = 0$ since all values in S_X are above $y = a$

For Region 3: $a \leq y < b$, considering $y_1 = g(x_1)$

$F_Y(y_1) = \Pr(g(X) \leq y_1) = \Pr(X \leq g^{-1}(y_1)) = \Pr(X \leq x_1) = F_X(x_1) = F_X(g^{-1}(y_1))$

Overall CDF of Y is

$$F_Y(y) = \begin{cases} 0, & y < a \\ F_X(g^{-1}(y)), & a \leq y < b \\ 1, & y \geq b \end{cases}$$

The PDF of Y can be found by taking the derivative of $F_Y(y)$

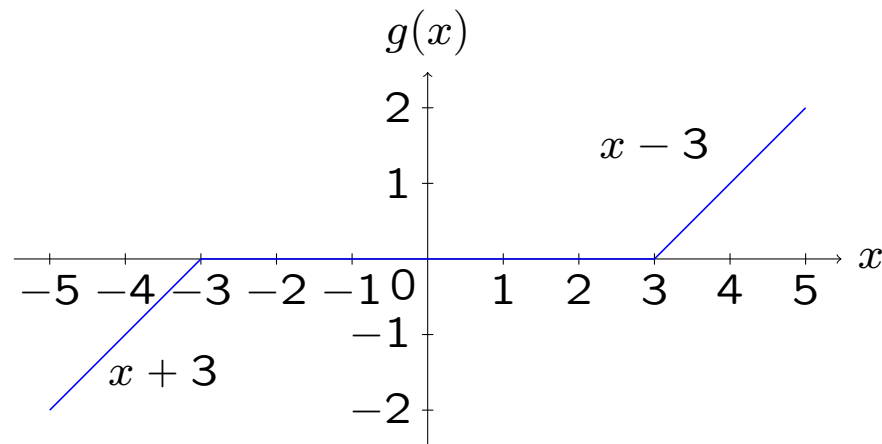
Case 3. Consider CRV X and $Y = g(X)$

if $g(x)$ is constant for a range of x values

$\Rightarrow Y$ is a mixed RV with a discontinuity at Y at that constant value

Example of Case 3: amplitude limiter **[Kay, Example 10.9];**

Example: Consider $Y = g(X)$ with $g(x) = \begin{cases} x - 3, & x > 3 \\ x + 3, & x < -3 \end{cases}$



Three regions to consider to find $F_Y(y) = \Pr(X \in R_y)$

$$R1: y > 0, F_Y(y) = F_X(y + 3)$$

$$R2: y < 0, F_Y(y) = F_X(y - 3)$$

$$R3: y = 0, \Pr(Y = 0) = \Pr(-3 < X < 3) = F_X(3) - F_X(-3)$$

Y is a mixed RV, and its CDF has a discontinuity at $Y = 0$ of size $F_X(3) - F_X(-3)$

If it is given that X is uniform between -3 and 3, then $X \sim \mathcal{U}(-3, 3)$ with CDF

$$F_X(x) = \begin{cases} 0, & x < -3 \\ (x + 3)/6, & -3 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

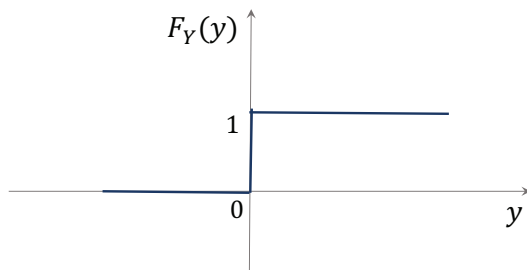
From $R1$: $y > 0$, only obtain $F_Y(y) = 1$, $y > 0$ since all other values are for $y < 0$,

$$F_Y(y) = F_X(y + 3) = \begin{cases} 0, & y < -6 \\ (y + 6)/6, & -6 \leq y < 0 \\ 1, & y \geq 0 \end{cases},$$

Similarly, from $R2$: $y < 0$, only obtain $F_Y(y) = 0$, $y < 0$

And from $R3$, discontinuity at $y = 0$ of value $\Pr(Y = 0) = F_X(3) - F_X(-3) = 1 - 0 = 1$

Combining regions, $F_Y(y) = \begin{cases} 0, & y < 0 \\ 1, & y \geq 0 \end{cases}$



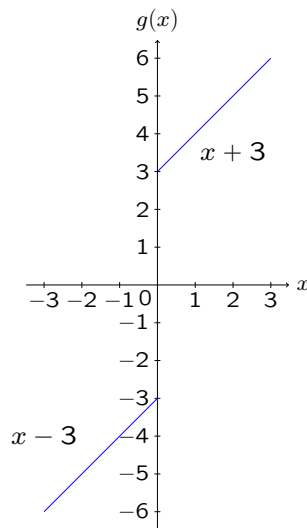
Case 4.

$Y = g(X)$, single value $x = x_0$ transforms to a range of y values, e.g., $y_1 < y < y_2$ in S_Y

$\Rightarrow F_Y(y) = F_X(x_0)$ is constant for $y_1 < y < y_2$

Example: $Y = g(X)$ with $g(x) = \begin{cases} x + 3, & x > 0 \\ x - 3, & x < 0 \end{cases}$

Three regions to consider to find $F_Y(y)$: $R1: y > 3$, $R2: y < -3$, and $R3: -3 \leq y \leq 3$



$$\text{Resulting CDF: } F_Y(y) = \begin{cases} F_X(y + 3), & y < -3 \\ \Pr(X \leq 0) = F_X(0), & -3 \leq y \leq 3 \\ F_X(y - 3), & y > 3 \end{cases}$$